

في هذا المشروع قمّت بإنشاء نموذج للتعرف على الصور باستخدام Google Teachable Machine، حيث أنشأت فئتين للتصنيف (Cat و Dog) وقمت بتدريب النموذج باستخدام صور لكل فئة، ثم اختبرت النموذج للتأكد من دقة التصنيف.

بعد الانتهاء من التدريب، قمّت بتصدير النموذج بصيغة Keras → TensorFlow، وتم الحصول على ملفات النموذج (keras_model.h5) وملف أسماء الفئات (labels.txt).

بعد ذلك استخدمت Google Colab لكتابة وتشغيل برنامج بلغة Python يقوم بتحميل النموذج، وقراءة صورة الإدخال، ثم التنبؤ بالفئة وعرض نسبة الثقة.

أظهر النموذج نتيجة صحيحة حيث صنّف الصورة على أنها Dog بنسبة ثقة بلغت ٩٩,٩٩٩٩٦٤٪.

Teachable Machine

5 Image Samples

Webcam Upload

cat

5 Image Samples

Webcam Upload

Add a class

Training

Model Trained

Advanced

Preview

Export Model

Google Drive

Output

dog

100%

cat

Tensorflow.js

Tensorflow

Tensorflow Lite

Model conversion type:

☒ Keras
 ☐ Savedmodel
 [Download my model](#)

Converts your model to a keras.h5 model. Note the conversion happens in the cloud, but your training data is not being uploaded, only your trained model.

Code snippets to use your model:

Keras

OpenCV Keras

Contribute on Github

```

from keras.models import load_model # TensorFlow is required for Keras to work
from PIL import Image, ImageOps # Install pillow instead of PIL
import numpy as np

# Disable scientific notation for clarity
np.set_printoptions(suppress=True)

# Load the model
model = load_model("keras_Model.h5", compile=False)

# Load the labels
class_names = open("labels.txt", "r").readlines()

# Points the array of the first class to feed into the keras model
          
```

Copy

Google Drive

100%

The image displays two sequential screenshots of a Google Colab notebook titled 'Untitled0.ipynb'. The interface includes a file explorer on the left, a code editor in the center, and a terminal at the bottom.

Top Screenshot: The code editor shows the initial setup for loading a Keras model. The code imports necessary libraries, loads the 'keras_model.h5' file, and reads the 'labels.txt' file to get class names. It also sets up the image path to 'images.jpg'.

```
[1] ✓ 10s
from keras.models import load_model # TensorFlow is required for Keras to work
from PIL import Image, ImageOps # Install pillow instead of PIL
import numpy as np
import tf_keras as tk
# Disable scientific notation for clarity
np.set_printoptions(suppress=True)

# Load the model
model = tk.models.load_model("keras_model.h5", compile=False)

# Load the labels
class_names = open("labels.txt", "r").readlines()

# Create the array of the right shape to feed into the keras model
# The 'length' or number of images you can put into the array is
# determined by the first position in the shape tuple, in this case 1
data = np.ndarray(shape=(1, 224, 224, 3), dtype=np.float32)

# Replace this with the path to your image
image = Image.open("images.jpg").convert("RGB")
```

Bottom Screenshot: The code editor shows the prediction step. It uses the loaded model to predict the class name and confidence score for the image. The output shows the prediction is 'dog' with a confidence score of 0.99999964.

```
[1] ✓ 10s
class_name = class_names[index]
confidence_score = prediction[0][index]

# Print prediction and confidence score
print("Class:", class_name[2:], end="")
print("Confidence Score:", confidence_score)

... 1/1 [=====] - 2s 2s/step
Class: dog
Confidence Score: 0.99999964
```